Weeks 4 and 5: Hypothesis testing

F79MA Statistical Models A

Heriot-Watt University

Hypothesis testing

In this unit we revisit the classical approach to hypothesis testing that was introduced in earlier statistical courses. We focus on the theoretical aspects of the topic and give a proof of the Neyman-Pearson Lemma, which is a key theoretical result in classical hypothesis testing.

Revision of classical approach to hypothesis testing

Let $\underline{X}$ be a random sample from a population with p.d.f (or p.m.f) $f_X(x;\theta)$. Here θ may be a single parameter, or a vector of parameters as appropriate.

A *hypothesis* is a statement about θ with regard to a subset of the parameter space Ω. For example, in the case where θ is one-dimensional, the hypothesis could be $\theta = 1$, or $\theta \geq 5$, or more generally $\theta \in \omega$, where ω is some subset of Ω.

Hypothesis tests typically involve two competing hypotheses: the *null hypothesis*

$$H_0 : \theta \in \omega\,,$$

and the *alternative hypothesis*

$$H_1 : \theta \in \omega' = \Omega \setminus \omega.$$

The null hypothesis represents a 'default' position, or the status quo, that we will only abandon (reject) if the data we collect give sufficient cause (evidence) to question it.

The value of θ is unknown, so we do not know which hypothesis is true. To conduct the hypothesis test, we observe some data $\underline{x}$, which we assume to be a realisation of a random variable $\underline{X}$ related to θ by a statistical model with probability density/mass function $f_{\underline{X}}(\underline{x};\theta)$, and use this information to assess the degree of support for H_0 relative to the support for H_1.

A hypothesis test is essentially a rule for partitioning the sample space (for the data $\underline{x}$) into two regions, as follows:

- A critical or rejection region, denoted R, such that if $\underline{x} \in R$ then we **reject** H_0,
- And its complement, R', such that if $\underline{x} \in R'$, then we **fail to reject** H_0.

Note that we generally cannot come to a firm conclusion on the basis of a hypothesis test so we write "fail to reject H_0" instead of "accept H_0". Furthermore, for the same reason if we reject H_0, this does not mean that we "accept H_1".

Hypothesis tests have limitations stemming from the uncertainty involved. In particular, there are two distinct types of error that we can make:

Type I error: A Type I error occurs when we incorrectly reject H_0, although it is actually true. The probability of a Type I error is

$$\alpha(\theta) = \mathbf{P}(\underline{X} \in R), \theta \in \omega.$$

Type II error: A Type II error occurs when we fail to reject H_0, although it is false. The probability of a Type II error is

$$\beta(\theta) = \mathbf{P}(\underline{X} \in R'), \theta \in \omega'.$$

A related concept is the function $1 - \beta(\theta)$, which is known as the *power* function of the test.

A good hypothesis test specifies R and R' in a way that minimises both α and β. Unfortunately, α and β are subject to a trade-off, and decreasing one type of error will typically increase the other. Designing an effective hypothesis test therefore involves a compromise between ensuring a small value of α and ensuring a small value of β.

Formulating hypothesis tests using the Neyman-Pearson Lemma

The standard starting point for designing a test is to first fix the value of α, or more generally to bound $\max\{\alpha(\theta)|\theta \in \omega\}$ if H_0 is not a simple or point hypothesis. Usually α is set to be small, e.g., 0.05 (5%) or 0.01 (1%). The chosen value of α is known as the *size* or *level* of the test.

Having fixed the value of α, we then try to find the test which minimises β, or equivalently, maximises the power $1–\beta$ subject to attaining the level α. This presents a challenge, namely:

How do we construct the optimal test, i.e. the one with the smallest β, for a given value of α?

For the case of two simple hypotheses, this can be straightforwardly addressed using the following very important statistical result:

The Neyman-Pearson Lemma (NPL) Suppose we have two simple hypotheses

$$H_0 : \theta = \theta_0 \; v. \; H_1 : \theta = \theta_1.$$

We wish to design a test of $H_0 \; v. \; H_1$ based on observation of a random sample $\underline{X}$. This can be specified by specifying the rejection region R for the test. The NP Lemma tells us that the most powerful (MP) test of size α is such that

$$R = \left\{ \underline{x} \mid \frac{f_{\underline{X}}(\underline{x}; \theta_0)}{f_{\underline{X}}(\underline{x}; \theta_1)} \leq k \right\},$$

where k depends on the size of the test. (The smaller α, the smaller k must be.)

Some notes:

1. Because the test is based on the likelihood ratio (LR) it respects the likelihood principle. Also, the only way in which the data $\underline{x}$ enter the test is through the likelihood $f(\underline{x};\theta)$.

2. If a sufficient statistic $s(\underline{x})$ exists for θ then the LR depends only on the value of $s(\underline{x})$. Recall from the factorisation theorem that

$$f(\underline{x};\theta) = h_1(\underline{x})h_2(s(\underline{x}),\theta).$$

It is therefore immediate that

$$\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)} = \frac{h_2(s(\underline{x}),\theta_0)}{h_2(s(\underline{x}),\theta_1)}.$$

Applying the Neyman-Pearson Lemma Applying the NP Lemma to construct optimal tests for two simple competing hypotheses involves the following steps

1. First we construct the ratio $\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)}$ and the associated critical region R.

2. We re-express R the form $\{\underline{x} : s(\underline{x}) \leq k^* \text{ (or } \geq k^*)\}$ where $s(\underline{x})$ is a function of $\underline{x}$ whose sampling distribution is known under H_0 and H_1.

3. We use the sampling distribution of $s(\underline{X})$ under H_0 to identify the required value of k^* to give the desired error rate α.

Example: Normal distribution - simple hypothesis. Let $\underline{X}$ be a random sample of size n from $N(\mu,\sigma^2)$ distribution where σ^2 is known. Test

$$H_0 \ : \ \mu = \mu_0 \ v. \ H_1 \ : \ \mu = \mu_1 \, .$$

Identify the reject region for $\alpha = 0.05$, $\mu_0 = 5$ and $\mu_1 = 6$ when $\sigma^2 = 1$, and $n = 10$. What is the power of this test?

For this model we have that

$$\begin{aligned}
L(\mu) &\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right), \\
R &= \{\underline{x} : \frac{L(\mu_0)}{L(\mu_1)} \leq k_1\} \\
\frac{L(\mu_0)}{L(\mu_1)} &= \exp\left(-\frac{1}{2\sigma^2}\left[\sum x_i^2 - 2\mu_0\sum x_i + n\mu_0^2\right] + \frac{1}{2\sigma^2}\left[\sum x_i^2 - 2\mu_1\sum x_i + n\mu_1^2\right]\right) \\
&= \exp\left(-\frac{1}{\sigma^2}(\mu_1-\mu_0)\sum x_i + \frac{1}{2\sigma^2}n(\mu_1^2-\mu_0^2)\right).
\end{aligned}$$

with k_1 set such that the type 1 error probability $\mathbf{P}(\underline{X} \in R) = 0.05$ when $\mu = \mu_0 = 5$.

Rather than deriving k_1, we use the fact that the likelihood ratio is a *monotone decreasing* function of $\sum x_i$ (since $\mu_1 > \mu_0$). Therefore, the region R may be equivalent defined as

$$R = \{\underline{x} \ : \ \sum x_i \geq k_2\}.$$

with k_2 set such that the type 1 error probability $\mathbf{P}(\underline{X} \in R) = 0.05$ when $\mu = \mu_0 = 5$ (recall that $\sum X_i$ is sufficient for μ). Moreover, we can also express R as

$$R = \{\underline{x} \ : \ \overline{x} \geq k_3\},$$

or

$$R = \{\underline{x} \ : \ \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}} \geq k_4\},$$

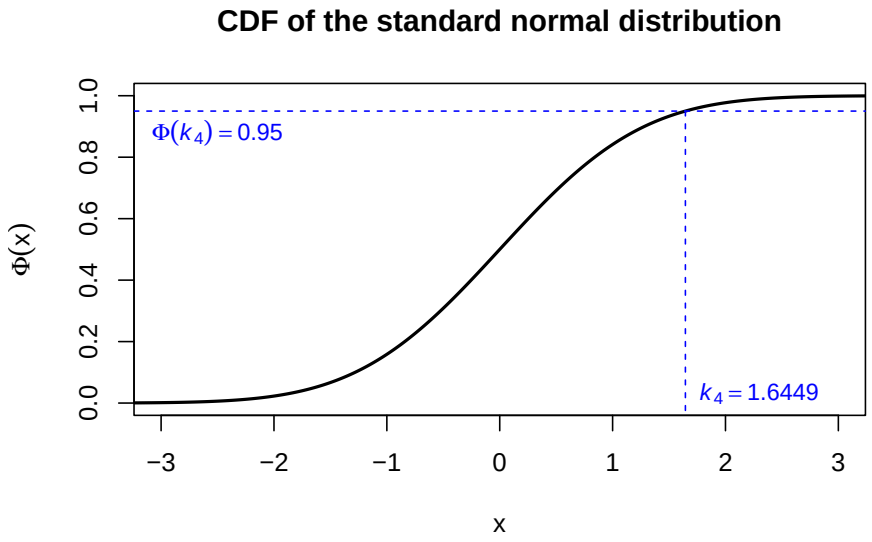
as identifying k_4 is easier than identifying k_2 or k_3. More precisely, when H_0 is true, we have the *sampling distribution*

$$\frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

to find that for $\alpha = 0.05$

$$k_4 = 1.6449.$$

A visualisation is as follows:



Then for $\mu_0 = 5, \sigma^2 = 1, n = 10$

$$k_3 = 5 + 1.6449\frac{1}{\sqrt{10}} = 5.52 \quad \text{and} \quad k_2 = 10k_3 = 55.2.$$

[**N.B.**: this has been determined without reference to the value of μ_1 (other than $\mu_1 > \mu_0$). The value μ_1 is required to determined the power of the test.]

The *Power of the test* is

$$1 - \mathbf{P}_{H_1}(\overline{X} \leq 5.52) = \mathbf{P}_{H_1}(\overline{X} \geq 5.52).$$

To determine the power of the test, we need a value from μ_1. For $\mu_1 = 6$,

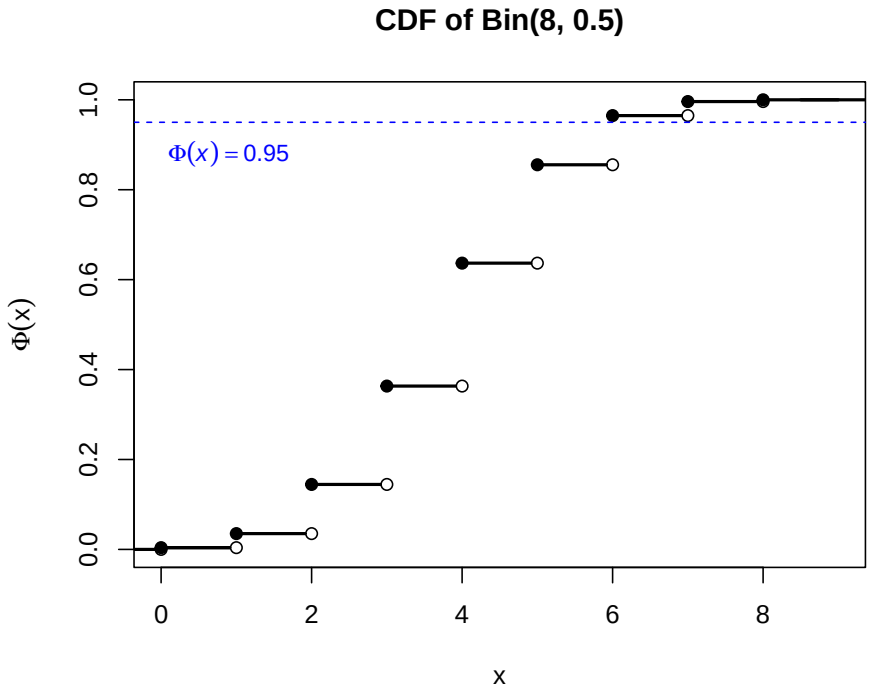
$$\text{Power} = \mathbf{P}\left(\frac{\overline{X} - 6}{1/\sqrt{10}} \geq \frac{5.52 - 6}{1/\sqrt{10}}\right) = \mathbf{P}(Z \geq -1.52) = 0.936.$$

Here $Z \sim N(0, 1)$.

Exercise: Let $X_1, .., X_n$ be a random sample from Exp(θ). Use the NP Lemma to find the most powerful test of $H_0 : \theta = 1$ vs. $H_1 : \theta = 2$ (of size α). For a given value of n how would you calculate the threshold for rejection and the power of the test?

Example: Discrete random variables In situations where discrete random variables are involved, it may not be possible to find an exact test of size $\alpha \in (0, 1)$. To recognise this, recall that for discrete random variables, the cumulative distribution functions (cdf) are not 'onto' the set of real numbers $(0, 1)$ since the cdfs are not absolutely continuous.

Consider a Binomial random variable $X \sim \text{Bin}(n, p)$, where $n = 8$. Suppose we are interested to test $H_0 : p = 0.5$ v. $H_1 : p = 0.7$. It can be shown that the critical region is $R = \{x : x \geq k\}$. To design a test with size $\alpha = 0.05$, we need to determine the value of k so that $\mathbf{P}(X \geq k|H_0) = 0.05$, or equivalently, $\mathbf{P}(X \leq \lceil k \rceil - 1|H_0) = 0.95$. The cdf of X under H_0 is a step function shown below:



It is clear that there is no value $x \in \mathbb{R}$ such that $\Phi(x) = 0.95$, i.e., we cannot find k such that the Type I error probability is exactly 5%. There are several ways to go about this:

(i) Design the test so that the Type I error probability is at most equal to 0.05. For instance, we can choose the value k that gives a Type I error probability that is just under 5%. From the graph, we can choose $k = 7$ which gives a Type I error probability of $\alpha = 0.03516$.

(ii) Provided that n is large enough, we can apply Normal approximation to the Binomial to obtain an approximate distribution of X, i.e., $X \approx \mathrm{N}(np, np(1-p))$. In this case, the value of k can be easily obtained using statistical tables or any statistical software. Continuity correction may be required since we are approximating a discrete distribution with a continuous distribution.

(iii) Use a *randomisation* technique. The formulation of a randomised hypothesis test is highly theoretical, hence we will avoid discussing it in the lectures. You will have the opportunity to work with one example in the tutorials.

Tests for composite hypotheses: 1-sided tests

We now study tests for composite hypotheses. In this case, we seek a test which maximises the power for all values of θ consistent with H_1, i.e. for all $\theta \in \omega'$. When a single test verifies the NPL for all $\theta \in \omega'$ we refer to this test as the Uniform Most Powerful Test (UMP). Not all testing problems admit an UMP.

For illustration, consider a random sample $X_1, .., X_n$ from E$xp(\theta)$. One can show that the most powerful test of $H_0 : \theta = 1$ $v.$ $H_1 : \theta = 2$ of size α has critical region

$$R = \{\underline{x} : \sum x_i \leq k\}.$$

Note that we would have derived the same test for any H_1 where the value of θ that was greater than 1. This means that the above test is a UMP of

$$H_0 : \theta = 1 \; v. \; H_1 : \theta > 1.$$

Under what circumstances can we expect to find UMP tests? For 1-sided tests of the form

$$H_0 : \theta = \theta_0 \; v. \; H_1 : \theta > \theta_0. \quad \text{or} \quad H_0 : \theta = \theta_0 \; v. \; H_1 : \theta < \theta_0 \,,$$

we will find that a UMP exists if the population distribution has the *Monotone Likelihood Ratio Property* (MLRP).

A family of pdf's or pmf's has the MLRP if there exists a statistic $g(\underline{X})$ such that for every $\theta_1 > \theta_0$, $\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)}$ is a strictly monotonic increasing function of g(x). Note that we just need monotonicity, since if monotonic decreasing we could simply replace $g(\underline{x})$ with $h(\underline{x}) = -g(\underline{x})$.

We say that a random variable has a distribution from the exponential family if we can write the probability density function or probability mass function in the form

$$f(x;\theta) = c(\theta)h(x)\exp\{q(\theta)t(x)\}$$

for some functions c, h, q and t. You will see such distributions covered in more detail in F79MB Statistical Models B if you take that course. Such random variables will have the MLR property so long as $q(\theta)$ is strictly monotonic so this provides a straightforward method of checking whether the population distribution has the MLRP.

Example: Normal distribution - composite hypothesis. Let $\underline{X}$ be a random sample of size n from $N(\mu, \sigma^2)$ distribution where σ^2 is known. Test

$$H_0 \ : \ \mu = \mu_0 \ v. \ H_1 \ : \ \mu > \mu_0 \, .$$

Identify the reject region for $\alpha = 0.05$ when $\mu_0 = 5$, $\sigma^2 = 1$, and $n = 10$.

We can repeat the argument exactly as for the simple hypothesis

$$H_0 \ : \ \mu = \mu_0 \ v. \ H_1 \ : \ \mu = \mu_1 \, .$$

Exercise: Show that the MLRP holds for the Exponential, Binomial, and Poisson models.

Probability-values or p-values

In the early stages of hypothesis testing, hypotheses were tested at a fixed level (value of α) and rejected or not at that given level. A more modern approach is to report a probability or p-value as a way of summarising the conclusions of an analysis and the degree of support for H_0. The p-value is the minimum level of test at which H_0 would be rejected given the observed data, or equivalently, as the probability of observing data 'as extreme as' the given data when H_0 is true.

Example: For the normal example discussed above, with

$$H_0 \ : \ \mu = \mu_0 = 5 \ v. \ H_1 \ : \ \mu > 5.$$

$\sigma^2 = 1$ (known), and $n = 10$, let us calculate the p-value for H_0 when $\bar{x} = 5.73$.

Under H_0, $\overline{X} \sim N(\mu_0, \sigma^2/n) = N(5, 1/10)$. Thus, denoting by Z a standard normal random variable, the p-value is

$$\begin{aligned}
\mathbf{P}(\overline{X} \geq \bar{x}) &= \mathbf{P}(\overline{X} \geq 5.73) \\
&= \mathbf{P}\left(\frac{\overline{X} - 5}{\sqrt{1/10}} \geq \frac{5.73 - 5}{\sqrt{1/10}} \right) \\
&= \mathbf{P}(Z \geq 0.73 * \sqrt{10}) \\
&= 1 - \mathbf{P}(Z \leq 2.308) \\
&= 1 - 0.990 = 0.01
\end{aligned}$$

We now note some important comments regarding p-values.

- A p-value is NOT Pr(H_0 true | the data). Note that p-values describe frequencies of events over repeated sampling under H_0.
- There is a natural connection between p-values and testing at fixed level α - if the p-value is less than α, then reject at level α.
- The p-value "quantifies the evidence" against H_0 - the smaller the p-value the stronger the evidence against H_0.

Generalised Likelihood Ratio Tests (GLRTs)

UMP tests do not exist when H_1 is 2-sided, e.g.:

$$H_0 : \theta = \theta_0 \; v. \; H_1 : \theta \neq \theta_0,$$

as the most powerful test (as specified by NP Lemma) would be different for different values of θ_1 depending on whether $\theta_1 > \theta_0$ or $\theta_1 < \theta_0$. Nevertheless, we can usually design 2-sided hypothesis tests with 'good' properties by adopting the generalised likelihood ratio testing approach described below.

Recall that the NP Lemma gives the best (MP) test for two simple hypotheses. It is based on the likelihood ratio $\frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)}$. The Generalised Likelihood Ratio Test (GLRT) extends this to composite hypotheses in a very natural way. Consider testing

$$H_0 : \theta \in \omega \; v. \; H_1 : \theta \in \omega'$$

The GLRT does this by considering the ratio $\frac{f_{\underline{X}}(\underline{x};\hat{\theta}_0)}{f_{\underline{X}}(\underline{x};\hat{\theta}_1)}$ where $\hat{\theta}_0$ and $\hat{\theta}_1$ are the values of θ that maximise the likelihood over ω and ω' respectively. In practice, when ω' is the complement of ω, it is sufficient (and usually simpler) to consider the unrestricted MLE $\hat{\theta}$ in the denominator and obtain a test with critical region

$$R = \{\underline{x} \mid \frac{f_{\underline{X}}(\underline{x};\hat{\theta}_0)}{f_{\underline{X}}(\underline{x};\hat{\theta})} \leq k\},$$

where k is chosen to give a test of the requisite size α.

The reason for this is that we reject H_0 when $f_{\underline{X}}(\underline{x};\hat{\theta}_0) \ll f_{\underline{X}}(\underline{x};\hat{\theta}_1)$, which can only happen if $\hat{\theta}_1 = \hat{\theta}$ (the *unrestricted* MLE). Therefore in cases where rejection could be sensible, the two ratios coincide. Lastly, we don't give examples here, but in 1-sided situations where the MLRP holds, the GLRT usually leads to the UMP test.

For clarity, we use the notation

$$\lambda(\underline{x}) = \frac{f_{\underline{X}}(\underline{x};\hat{\theta}_0)}{f_{\underline{X}}(\underline{x};\hat{\theta})}.$$

In designing a test with a desired Type I error rate α it will be important to understand the sampling distribution of $\lambda(\underline{X})$ under H_0 so that an appropriate threshold can be determined. In general this will be difficult to do but in the large-sample setting there are asymptotic results which we can use to approximate the distribution of $\lambda(\underline{X})$ under H_0.

Asymptotic Result for the GLRT

Recall the asymptotic results for the distribution of the MLE and of the deviance. We can find a similar result for the distribution of $\lambda(\underline{X})$ under H_0 as $n \to \infty$, and use it as an approximation for large n. Specifically, for sufficiently large n we have (approximately):

$$T(\underline{X}) = -2\log\lambda(\underline{X}) \sim \chi^2_d,$$

where d is the difference in dimensionality between the parameter space Ω and the dimensionality of the set ω. This asymptotic result can be used to calculate the appropriate value of k in the GLRT to give any desired value of α, up to some approximation error that we can evaluate analytically or by using simulation in R.

Note that $d = 1$ in many of our examples.

Example: N(μ, σ^2) with σ^2 known.

$$H_0 : \ \mu = \mu_0 \ v. \ H_1 : \mu \neq \mu_0.$$

Here $d = 1$, since ω is actually a point in a 1-dimensional parameter space, so that $T(\underline{X}) = -2\log\lambda(\underline{X}) \sim \chi^2_1$ (The result is *exact* in this case for all n.)

If σ^2 is also unknown then for

$$H_0 : (\mu, \sigma^2) = (\mu_0, \sigma_0^2) \ v. \ H_1 : (\mu, \sigma^2) \neq (\mu_0, \sigma_0^2)$$

$d = 2$, so that $T(\underline{X}) = -2\log\lambda(\underline{X}) \sim \chi^2_2$.

More challenging material: outline proof of the NP lemma

Consider the critical region specified by the NP Lemma

$$R = \left\{ \underline{x} \mid \frac{f_{\underline{X}}(\underline{x};\theta_0)}{f_{\underline{X}}(\underline{x};\theta_1)} \leq k \right\},$$

so that R describes a subset of the sample space of the $\underline{X}$. This specifies a test (call it Test 1) with Type I error rate α (controlled by the choice of k), and consequent Type II error rate β.

Now consider an arbitrary alternative test, Test 2, with critical region R_2. Suppose we have designed Test 2 to have the the same Type I error rate as Test 1, namely α. We will show that the Type II error rate of Test 2 must be at least β.
To do this we will focus on two regions

A: Points in R which are not in R_2.

B: Points in R_2 which are not in R.

Assuming that H_0 is true, an experimental outcome in A will cause a Type I error under Test 1 but not under Test 2, while one in B will cause a Type I error under Test 2 but not under Test 1. Since both tests have same value of α it follows that

$$\int_A f_{\underline{X}}(\underline{x}, \theta_0) d\underline{x} = \int_B f_{\underline{X}}(\underline{x}, \theta_0) d\underline{x}.$$

We will now show that Test 2 must have a higher Type II error rate than Test 1. Let β_2 be the Type II error rate of Test 2. Remember that Type II errors relate to events under the alternative hypothesis so that we must use θ_1 in the calculations. It follows that

$$\beta_2 = \beta + \int_A f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} - \int_B f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x}$$

On A,

$$\frac{f_{\underline{X}}(\underline{x}; \theta_0)}{f_{\underline{X}}(\underline{x}; \theta_1)} \leq k$$

so that $f_{\underline{X}}(\underline{x}, \theta_1) \geq f_{\underline{X}}(\underline{x}, \theta_0)/k$. It follows that

$$\int_A f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} \geq \int_A f_{\underline{X}}(\underline{x}, \theta_0) d\underline{x}/k = \Delta.$$

On B,

$$\frac{f_{\underline{X}}(\underline{x}; \theta_0)}{f_{\underline{X}}(\underline{x}; \theta_1)} > k$$

so that $f_{\underline{X}}(\underline{x}, \theta_1) < f_{\underline{X}}(\underline{x}, \theta_0)/k$. It follows that

$$\int_B f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} < \int_B f_{\underline{X}}(\underline{x}, \theta_0) d\underline{x}/k = \int_A f_{\underline{X}}(\underline{x}, \theta_0) d\underline{x}/k = \Delta.$$

It follows that

$$\int_A f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} > \int_B f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x}$$

so that

$$\beta_2 = \beta + \int_A f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} - \int_B f_{\underline{X}}(\underline{x}, \theta_1) d\underline{x} > \beta$$

Therefore Test 2 must have at least as large a Type II error rate as Test 1. Note that the above construction is general. For any alternative Test 2 there will in general be regions A and B as defined above and the analysis above could be applied. This means we can't find any test with Type I error rate α that has a lower Type II error rate than β, so that Test 1 is the most powerful test of size α of H_0 v H_1. This completes our proof.